COSC2507 Project Proposal

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Statistical Anomaly Detection System for Fraudulent Banking Transactions

1. Project Summary

This initiative proposes a system for detecting anomalies to recognize potentially fraudulent transactions in banking. The primary method to be used will be a statistical framework based on the ImDiffusion concept introduced by (Zhang, n.d.). Instead of solely relying on direct classification, the proposed system will aim to detect an anomaly using z-score and Interquartile Range (IQR) techniques.

2. Background

Banks process a high volume of electronic transactions, including credit card payments, money transfers, and ATM withdrawals. Though fraudulent occurrences among transactions are relatively rare, even a few fraudulent cases can cause financial losses for both customers and financial institutions. Fraud detection is becoming increasingly challenging as fraudulent transactions can often be difficult to distinguish from legitimate transaction patterns, and new fraud patterns can emerge and spread rapidly as AI models evolve.

Statistical anomaly detection presents an alternate approach; instead of focusing on the characteristics of known fraudulent transactions, an anomaly detection system learns the behavior of normal transactions and detects those that significantly deviate from the norm.

3. **Research Question**:

The project will investigate the following research questions:

- How effective can the proposed approach detect anomalous transactions when the dataset involved is very imbalanced?

4. **Proposal Methodology**:

4.1 **Overview**

The project will adopt a quantitative research approach involving the collection of data. Preprocessing, exploratory data analysis, statistical anomaly detection, and a performance evaluation. The research will have the following stages:

- Data collection of a publicly available credit card fraud dataset.
- Data preprocessing and cleaning to prepare the dataset for analysis.
- Exploratory Data Analysis to investigate patterns in transaction data.
- Application of statistical anomaly detection techniques including Z-score and IQR.
- Evaluation of the proposed techniques using labelled fraud data.
- Interpretation of findings, limitations, and applications.

4.2 **Dataset**

The primary dataset that will be used for this project is the Credit Card Fraud Detection Dataset that is publicly available on Kaggle. This dataset comprises European cardholder transactions collected over two days in September 2013. The dataset includes:

- There are a total of 284,807 credit-card transactions, including 492 fraudulent transactions and 284,315 legitimate transactions.
- Transaction time.
- Transaction amount.
- A Classification variable where 0 denotes a legitimate transaction, and 1 denotes fraud.

The following tools and resources that would be used are as follows:

This project is mainly going to be a Python-based project.

- Python: Primary programming language for data processing and statistical analysis.
- Pandas: Data cleaning and manipulation.
- NumPy: Numerical computations and statistical calculations.
- Matplotlib and Seaborn: Data visualization and exploratory analysis.
- Scikit-learn: Data preprocessing and evaluation metrics.
- Jupyter Notebook: Environment for conducting experiments and documenting results.

- Kaggle: Source of the publicly available dataset.

4.2 **Planned technique**

The project will investigate two primary statistical anomaly detection techniques: Z-score analysis and the Interquartile Range. These techniques will be used to identify transactions that exhibit unusual numerical characteristics compared with the observed transaction data.

Z-score: This analysis is a statistical technique used to describe how many standard deviations a data point is below or above the mean of a given distribution.

The Z-score is calculated using the following formula:

$$Z = (x - \mu) / \Sigma$$

Where:

- Z represents the Z-score.
- x represents the observed transaction value.
- μ represents the mean of the feature.
- Σ sigma represents the standard deviation of the feature.

A transaction with an absolute Z-score greater than a selected threshold, e.g. 3, would be considered statistically unusual.

Interquartile Range (IQR): This technique is used to identify outliers based on the distribution of the dataset.

IQR is calculated by:

$$\mathbf{IQR = Q3 - Q1}$$

Where:

Q1 represents the first quartile.

Q3 represents the third quartile.

IQR represents the difference between the two quartiles.

The lower and upper boundaries for identifying potential outliers are calculated by:

Lower Bound= Q1 − 1.5(IQR)

Upper Bound= Q3 + 1.5(IQR)

Transactions with values outside of these bounds will be considered potential statistical outliers.

4.3 **Proposed Contributions**

The contributions expected from this project include:

- **Application of statistical techniques to fraud detection**: Investigating how Z-score and IQR methods can be used to identify unusual financial transactions.

- **Exploratory analysis of financial data**: Looking at transaction distributions and outliers to gain insights into fraudulent and legitimate transaction behaviour.

5. **Expected Outcome and Applications**

The project is expected to give insight into the potential effectiveness of statistical anomaly detection techniques for detecting suspicious banking transactions. As well as satisfying the following outcomes:

- Give a proposed framework for using statistical anomaly detection to support financial fraud monitoring.
- Assessment of Z-score and IQR methods for detecting potential anomalies.
- Use of comprehensive exploratory analysis of a publicly available credit card fraud dataset.
- Identification of unusual transaction patterns using statistical analysis.

An application of this project includes:

- **Fraud Analyst Decision Support**: Anomaly scores and statistical indicators could help fraud analysts prioritize transactions for manual review.
- **Exploratory Financial Data Analysis**: Statistical analysis could help financial institutions better understand transaction distributions, identify unusual patterns, and investigate changes in transaction behavior.

6. **Challenges**

Some challenges could be faced while building out this project:

- **Limitations of Statistical Methods**: Z-score and IQR methods primarily identify unusual numerical values and may not capture complex relationships among transaction features.
- **Limited Dataset Representation**: The selected public dataset covers only a short observation period and contains anonymized features.
- **Privacy Considerations**: Though this is not a project that involves real users, it is of best practice to follow proper data privacy ethics.

7. **Conclusion**

In conclusion, this project brings forth the idea of a statistical anomaly detection framework for detecting potentially fraudulent banking transactions. The research will investigate the application of Z-score analysis and the Interquartile Range to identify unusual transaction patterns in a publicly available credit card fraud dataset. Exploratory Data Analysis will be used to examine transaction distributions, identify outliers, and investigate the characteristics of fraudulent and legitimate transactions.

Ultimately, the project aims to demonstrate the effectiveness of statistical analysis in the field of financial fraud detection and its application for a real-world financial problem.

8. References

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